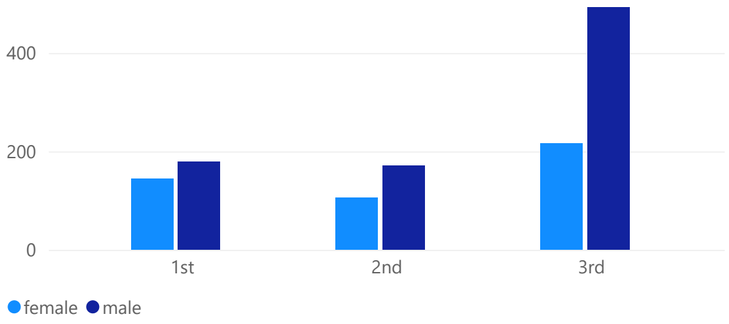
Data Analysis Results — Survival on the *Titanic*

Sex: All | Age Category: All | Class of Travel: All | Clear all slicers

Number of Passengers	Number of Survivors	Survival Rate
1309	497	38,41%

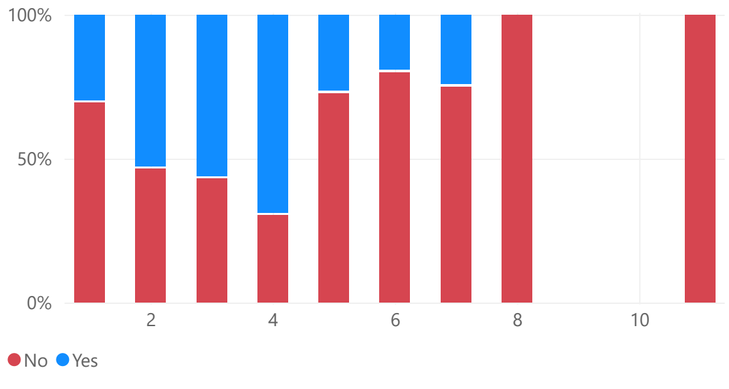
Number of Passengers
by Class and Sex



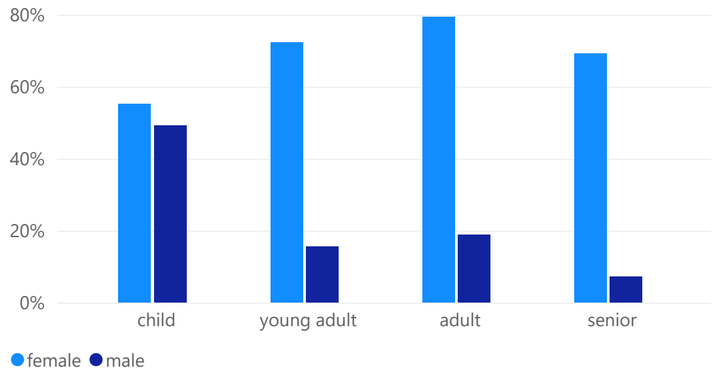
Survival Rate
by Class and Sex

class	female	male	Total
3rd	49,07%	15,10%	**25,43%**
2nd	88,68%	15,06%	**43,75%**
1st	96,50%	34,86%	**62,58%**
Total	**72,79%**	**19,25%**	**38,41%**

Survival Rate
by Family Size



Survival Rate
by Age Group and Sex



The Key Takeaways:

- Sex emerged as the single most important factor associated with survival. Women were considerably more likely to survive than men.
- The second most important factor was class of travel. However, its effect was particularly pronounced among men in first class and among women, while survival rates for men in second and third class were relatively similar.
- Other factors associated with survival included family size and age. Passengers travelling in smaller family groups generally had higher survival rates, while fare was strongly related to class of travel. Age also played a role, with younger passengers generally having higher survival rates. However, these factors had a much smaller impact than sex and class.

Model Performance

- Most models reach a similar accuracy level, around 80%-81%. Another common pattern is the difference between high Specificity and more moderate Sensitivity. The models tend to classify non-survivors more reliably than survivors.
- Based on the cross-validation results, the Random Forest model was selected as the final model, as it achieved the highest ROC-AUC among all evaluated models.
- The modeling results suggest that the dataset has a natural accuracy ceiling of approximately 80-82%, likely reflecting the noise and uncertainty in the data. This is unsurprising, as survival in a maritime disaster such as the Titanic was influenced by numerous factors that cannot be fully captured by the available variables.
- The main limitation of the analysis is the large amount of missing information. One of the potentially most informative variables is "Cabin", which contains information about the deck on which a passenger's cabin was located. This variable was excluded from the analysis because most of its values are missing.
- The main variable influencing the RF model is sex, which corresponds to the finding of the earlier analysis.

Model Metrics
Evaluated on a Training Subset

Model	ROC-AUC	Accuracy	Sensitivity	Specificity
Decision Tree	0,83	81,18%	68,09%	89,34%
k-Nearest Neighbours	0,84	78,57%	64,32%	87,46%
Logistic Regression	0,85	80,02%	71,61%	85,27%
Random Forest	0,87	80,21%	65,58%	89,34%
SVM (RBF)	0,85	80,21%	63,82%	90,44%

Random Forest Performance
Test set vs Training Set

Evaluation	ROC-AUC	Accuracy	Sensitivity	Specificity
Test Set	0,85	80,62%	62,63%	91,82%
Training Set (10-fold CV)	0,87	80,21%	65,58%	89,34%

Feature Importance
Random Forest - Training Subset

